

Aziforms - Requirements Document

Purpose: To define clear, comprehensive functional and non-functional requirements for a SaaS form creation, publishing, and response management application. This document is used as a reference for developers using code automation tools (e.g., GitHub Copilot) and for QA to generate automated unit tests and functional test cases.

Key Features & Objectives

- Users can create customizable forms with various field types.
- Forms can be published and shared via unique links.
- Response collection and storage with real-time statistics.
- Dashboard to view and analyze form responses.
- Form creators can export responses in various formats.
- Access control ensures only form creators can edit their forms.
- Automated notifications for new responses.
- Test coverage using code automation tools for both development and QA workflows.

User Roles

Role	Description
User	Can create forms, share links, and view responses.
Admin	Manages the platform, users, and has access to system analytics.

Access Control (Form-Based Attribute)

- Users authenticate via email with OTP verification.
- Only form creators can edit their own forms.
- Anyone with the form link can fill out and submit the form.

System Entities

Entity	Attributes
User	ID, Name, Email, Role, Creation Date
Form	ID, Title, Description, Creator ID, Creation Date, Status (Draft/Published), Access Type (Public/Private/Password-protected)
Form Field	ID, Form ID, Type (Text, Number, Choice, etc.), Label, Required Flag, Options (for choices), Validation Rules

Entity	Attributes
Response	ID, Form ID, Submission Date, Respondent Info (if collected)
Response Data	ID, Response ID, Field ID, Value

Functional Requirements

1. User Registration & Authentication

- Users register and login using email.
- Authentication via OTP (One-Time Password) sent to email.
- User profile management.

2. Form Creation

- Simple form builder interface.
- Multiple field types: text, number, email, phone, multiple choice, checkbox, dropdown, date, time, file upload.
- Field validation rules and conditional logic.
- Form theming and customization options.
- Save forms as drafts before publishing.

3. Form Publishing

- Generate shareable links for forms.
- Schedule form availability (start/end dates).

4. Response Collection

- Real-time response storage.
- Option to allow anonymous responses.
- Prevent multiple submissions from same user (optional).
- Response validation before submission.

5. Response Management

- Dashboard showing response statistics.
- Filter and search through responses.
- Export responses to CSV, Excel, or PDF.
- Automated data visualization for responses.

6. Notification System

- Email notifications for new responses.
- Notification settings customization.
- Summary reports (daily, weekly, monthly).

7. Form Access Control

- Only form creators can edit their forms.

- Anyone with the form link can submit responses.

QA-Centric User Stories (For Automation)

These user stories are designed to be testable using code automation tools.

User Story 1: Creating and Publishing a Form

As a User

I want to create a form and publish it

So that I can collect responses from my target audience.

Acceptance Criteria:

- Given I am logged in
- When I create a new form with multiple field types
- And I customize the form theme
- And I publish the form
- Then a unique link should be generated
- And the form status should change to “Published”
- And I should receive a confirmation notification

User Story 2: Form Access Restriction

As a Form Creator

I want to restrict access to my form with a password

So that only authorized individuals can submit responses.

Acceptance Criteria:

- Given I have a published form
- When I set access type to “Password-protected” and set a password
- Then anyone accessing the form should be prompted for the password
- And incorrect password attempts should be rejected
- And correct password entry should grant access to the form

User Story 3: Viewing Response Statistics

As a Form Creator

I want to view statistics about form responses

So that I can analyze the collected data.

Acceptance Criteria:

- Given my form has received multiple responses
- When I navigate to the form’s response dashboard
- Then I should see total response count

- And I should see graphical representations of response data
- And I should see completion rate statistics

User Story 4: Exporting Response Data

As a Form Creator

I want to export collected responses to a CSV file

So that I can analyze the data in external tools.

Acceptance Criteria:

- Given my form has received responses
- When I select the export option and choose CSV format
- Then a CSV file should be generated
- And the file should contain all response data
- And the file should be properly formatted for data analysis

User Story 5: Response Notifications

As a Form Creator

I want to receive notifications when my form receives new responses

So that I can stay updated on submission activity.

Acceptance Criteria:

- Given I have enabled notifications for my form
- When a new response is submitted
- Then I should receive an email notification
- And the notification should include basic response information
- And the notification should contain a link to view the full response

User Story 6: Access Control Enforcement

As a User

I want to ensure only I can edit my forms

So that their integrity is maintained.

Acceptance Criteria:

- Given I have created a form
- When another user tries to access the form's edit page
- Then they should be denied access
- And they should only be able to view and submit the form
- And only I should be able to edit or delete the form

User Story 7: OTP Authentication

As a User

I want to log in using my email with an OTP

So that I can securely access my forms without remembering a password.

Acceptance Criteria:

- Given I am on the login page
- When I enter my email address
- Then I should receive an OTP to my email
- And I should be able to enter the OTP to gain access
- And the OTP should expire after a short period of time